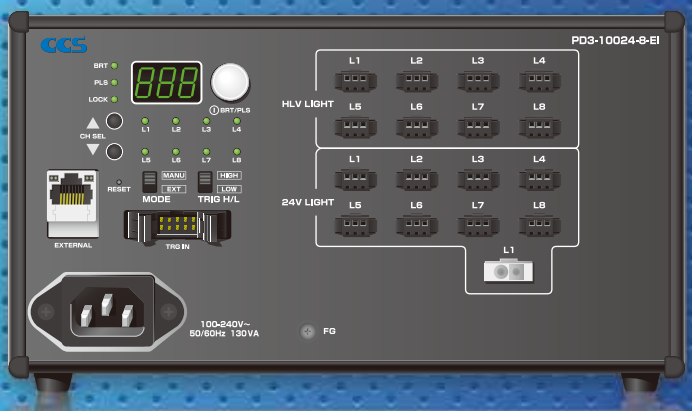
Control Unit for LED Light Units

PD3-10024-8-EI(A)

With Ethernet communications

Instruction Guide

Thank you for purchasing a CCS product. To ensure proper use of the product, please read this Instruction Guide before use and keep it for your future reference.



This Control Unit is specifically designed to control the light intensity of CCS LED Light Units. It is mainly used to control LED Light Units that are used for machine vision and industrial inspections.

Features

- Connect 24V DC Light Units and Spotlights. Use up to 8 channels.
- The light intensity can be manually controlled with a switch on the front panel, or externally controlled using Ethernet.
- Use PWM control to control the 24V DC Light Unit output at a frequency of 125 kHz.
- TCP/IP and UDP/IP Ethernet communications can be used for external control.
- Use external trigger inputs to turn Light Units ON and OFF, or to flash the strobe (for 24V DC Light Units only).

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1 Important Information for Equipment Safety — Read before Use

This product has been designed with full consideration of safety. Incorrect usage of the product may result in fire, electric shock, or other serious damages. Observe the following precautions.

The following symbols are used in this instruction guide to indicate and classify the relative importance of warnings and cautions.

⚠ WARNING	Indicates that incorrect usage may result in serious injury or death.	⚠ Caution	Indicates that incorrect usage may result in injury or property damage.

The following symbols in the instruction guide indicate and classify the precautions.

PROHIBITED, DISASSEMBLY PROHIBITED, DO NOT TOUCH WITH WET HANDS, DO NOT SUBJECT TO MOISTURE	INSTRUCTED, UNPLUG
These symbols mean the prohibited action.	These symbols indicate actions that must be performed.

⚠ WARNING

Do not disassemble or modify the Control Unit. Doing so may result in fire or electric shock. (DISASSEMBLY PROHIBITED)	Do not touch the plugs or switches with wet hands. Doing so may result in electric shock. (DO NOT TOUCH WITH WET HANDS)
Make sure that the Control Unit is free of moisture or any liquid. Doing so may result in fire or electric shock. (DO NOT SUBJECT TO MOISTURE)	Before connecting or disconnecting cables, make sure that the power source is turned OFF. Not doing so may result in fire or electric shock. (INSTRUCTED)
Do not touch the power cords during lightning. This may result in electric shock. (PROHIBITED)	If an abnormal condition occurs, such as fuming, heat, smell, or noise, stop using the Control Unit immediately, turn OFF the power source and unplug the power cord. Not doing so may result in fire or electric shock. (UNPLUG)

⚠ Caution

Do not connect any Light Units other than CCS LED Light Units. Doing so may cause overcurrent and the device may overheat or ignite. (PROHIBITED)	Always use one of the following power cords. 100 to 120 V range: SVT or SJT, AWG18, length: 3 m max., dielectric strength: 125 V min. 200 to 240 V range: H05VV-F, AWG18, length: 3 m max., dielectric strength: 250 V min. (INSTRUCTED)
Do not use user-made light cables. Doing so may cause Control Unit failure. (PROHIBITED)	Plug the power cord directly into a wall socket. Using a power strip or connecting many loads from one electrical outlet may cause fire or electric shock. (INSTRUCTED)
Do not place the Control Unit in direct sunlight or in a high-humidity environment. Doing so may result in fire due to internal temperature rise. (PROHIBITED)	Do not bundle Control Unit cables with high-voltage lines or power lines. Allow leeway when installing the cables. (INSTRUCTED)
Always place the Control Unit on a stable and flat location. Not doing so may result in the Control Unit falling or toppling, which may cause malfunction, accidents, or bodily injury. (PROHIBITED)	Always ground the power cord. Not doing so may cause Control Unit failure due to static electricity destroying electrical components including those in the Light Unit. (INSTRUCTED)
Do not drop the Control Unit or subject it to impact. Doing so may cause Control Unit failure. (PROHIBITED)	Use Light Units that are suitable for the Control Unit ratings. Exceeding the ratings may cause Control Unit failure. (INSTRUCTED)
Do not bend cables or jam them between objects when wiring. Doing so may cause Control Unit failure. (PROHIBITED)	Use a standard Extension Cable that is manufactured by CCS. However, if the cable is too long, the light intensity will decrease due to voltage drop caused by the DC resistance of the cable. (INSTRUCTED)
Do not intentionally short-circuit the positive and negative output terminals. (PROHIBITED)	Do not disconnect the power cord or disassemble the Control Unit during operation. Pulling on the cable may damage the cable and result in fire or electric shock. (INSTRUCTED)
Do not wipe the Control Unit with volatiles such as paint thinner or benzene. Discoloration or deterioration of the Control Unit surfaces may occur. (PROHIBITED)	Before moving the Control Unit, disconnect all connection cables. Damaging the cables may result in fire or electric shock. (INSTRUCTED)
Use a dry cloth to remove dust or other foreign matter from the electrodes. Failure to do so may result in fire. (INSTRUCTED)	When mounting the Control Units in system racks or cases, do not insert the screws more than 5 mm. Doing so may cause short-circuits in internal components. (INSTRUCTED)

2 Names and Functions of Parts

Setting Indicators
BRT lit: The light intensity can be set.
PLS lit: The lighting mode can be set.
LOCK lit: The settings are locked.

Digital Display
Displays the setting of the light intensity or the setting of the lighting mode.

Setting Switch
Press: Switches between the light intensity setting and lighting mode setting.
Press for at least 2 seconds: Locks the settings.
Rotate: Sets the light intensity or lighting mode.

Channel Selection Switch
Select the channel from L1 to L8. Only the channel on which Light Units are connected can be selected. The channels with same channel No. of 24V LIGHT and HLV LIGHT are operated simultaneously.
* It is not recommended to use the channels with same channel No. of 24V LIGHT and HLV LIGHT at the same time.

Channel Indicators
The indicator for the selected channel will light. When the L1 indicator is lit, the settings for the L1 Light Unit in the 24V Light Units (24V LIGHT) and the L1 Light Unit in the Spotlights (HLV LIGHT) can be changed.

External Control Reset Switch
Pressed with a pointed object to reset network settings to their default values.

External Control Connector
Used for external control with Ethernet communications.

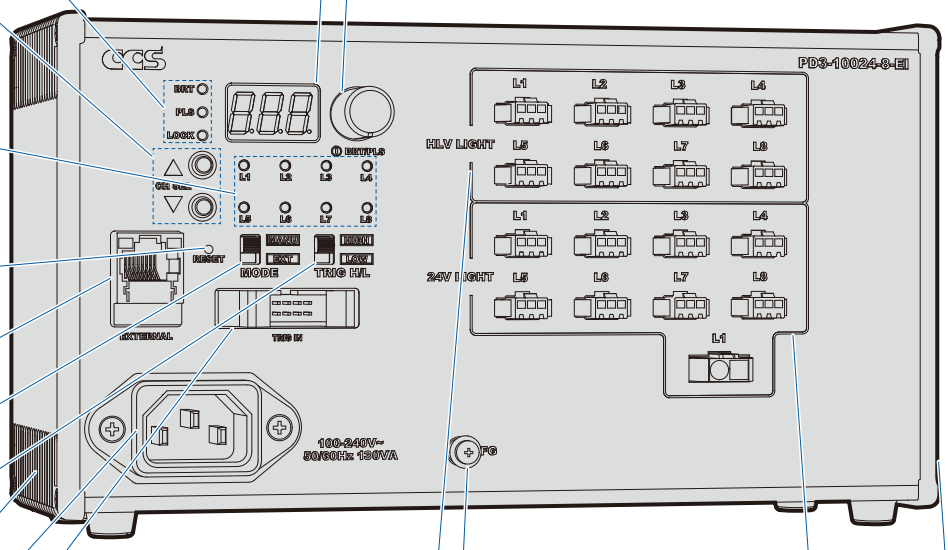
Manual/External Mode Selector
Selects manual (MANU) or external (EXT) control mode.

Trigger Logic Switch
Selects the logic of the trigger signal.

Fan Air Inlet (Left side)
This is the air inlet for the cooling fan.

AC Inlet
Connects the power source to the Control Unit.

External Trigger Input Connector
Inputs the ON/OFF signal for ON/OFF Mode.
Inputs the trigger signal for Strobe Mode.



Output Connectors (HLV LIGHT)
Connect these connectors to the Spotlights.
L1 to L8 (SMP-03V-BC)

FG Terminal
Ground the FG terminal.

Output Connectors (24V LIGHT)
Connect these connectors to the 24V DC Light Units.
L1 to L8 (SMP-03V-BC), L1(ELR-02V)

Fan Exhaust Outlet (Right side)
This is the air outlet for the cooling fan.

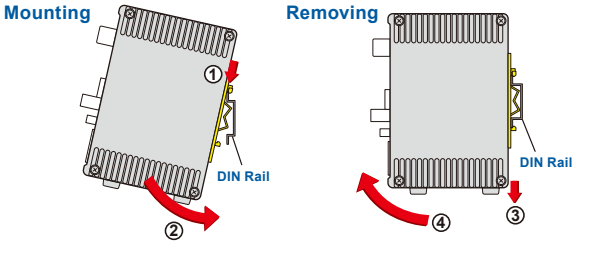
3 Installation

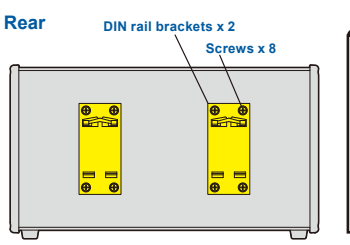
⚠ WARNING Do not place any objects within 20mm from the fan air inlets or fan exhaust outlets. Insufficient ventilation may cause heat to accumulate inside the product and result in a fire.

Mounting the Unit to DIN Rail

Mounting to DIN Rail
Hook the tab on the upper part of the Unit on the DIN rail and press the Unit in the direction indicated by arrow 2 while pressing it in the direction indicated by arrow 1.

Removing from DIN Rail
Press the Unit down in the direction indicated by arrow 3 and pull it out in the direction indicated by arrow 4.





⚠ Caution
Do not remove the DIN rail brackets. If they must be removed and attached again, make sure that you use the original screws (or M3 × 4 mm screws).
If other screws are used, they may short-circuit internal components and electric shock may occur.

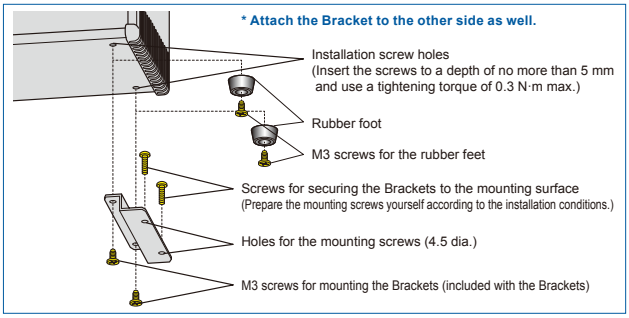
Securing the Unit with Base Brackets (Accessories)

⚠ Caution Always use Base Brackets (model: BK-PD3) when securing the Unit at its base. If it is secured without the Brackets, the Unit may be damaged.

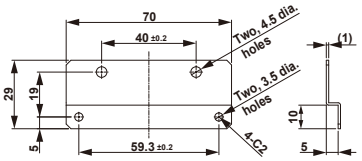
1 Removing the Rubber Feet from the Bottom of the Unit
Remove the screws that hold the rubber feet in place using a Phillips screwdriver.

2 Securing the Brackets to the Base of the Unit
Secure the Brackets to the base of the Unit with the four screws that come with the Brackets.

3 Securing the Unit with Mounting Screws
Secure the Unit in place with mounting screws. The mounting screws must be provided by the user.



Base Brackets Model: BK-PD3



Includes two Base Brackets and four mounting screws (+ one spare screw)

4 Connections

WARNING Before connecting the Control Unit, make sure that the main power source is turned OFF. Making connections with the power turned ON may result in a fire or electric shock.

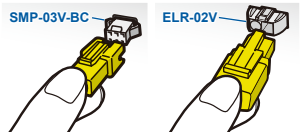
1 Connect the LED Light

Connect the connecting cable to the LED Light Unit to the output connector on the Control Unit.

Connect 24V DC Light Units to the 24V LIGHT output connectors, and Spotlights to the HLV LIGHT output connectors. One of the 24V LIGHT output connectors is an ELR-02V connector for a High-output Light Unit.

Connecting

Insert the connector to the Light Unit all the way in.



Removing

Press the lock and pull out the connector.



2 Connect the AC Cord

Connect the power cord to the Control Unit and a wall socket. The Control Unit will turn ON when power is supplied from the main power source. When the Unit is ON, the digital display will light.

Digital Display

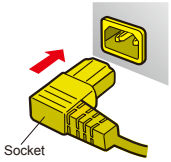
The light intensity of the lowest channel is displayed.



(Display when no Light Unit is connected)



*Data that has been set is retained even after the power is turned OFF with manual or external control.



* If you would like to use the Control Unit with 200 to 240 VAC, you must procure appropriate AC power cord.

Recommended specifications

Wire diameter	**1.0 mm² min.**
Line-to-line insulation withstand voltage	**100 MΩ min.**
Ratings	**250 V min., 10 A min.**
Withstand voltage	**2,000 VAC/minute**
Socket standards	**EN 60320-1 certified C13 type**

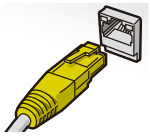
3 Connect the LAN Cable (Provided by the user)

To perform external control, connect the LAN cable to the external control connector.

The LAN cable must be provided by the customer. (Cable length should be within 30 m at aximum.) This product recognizes either cross-cable or straight-cable automatically by auto-negotiation function. Please reboot the product when changing the cable.

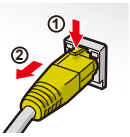
Connecting

Firmly insert the connector of the LAN cable.



Removing

Press the lock and pull out the connector.



4 Connect the External Trigger Input Cable (Optional item, sold separately)

To use ON/OFF Mode or Strobe Mode, connect the external trigger input cable to the external trigger input connector.

An optional External Trigger Input Cable (EXCB2-M10-3) (sold separately) is available. In case using a self-made cable, cable length should be within 3 m at maximum.

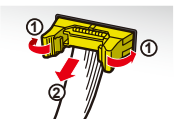
Connecting

Press the connector in until it locks in place.



Removing

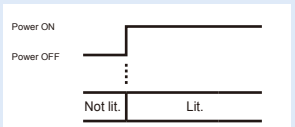
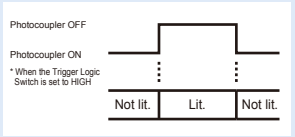
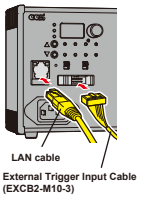
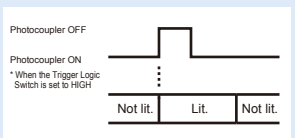
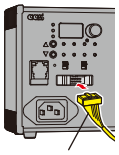
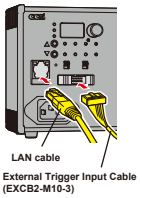
Release the lock and remove the connector.



5 What You Can Achieve with This Control Unit

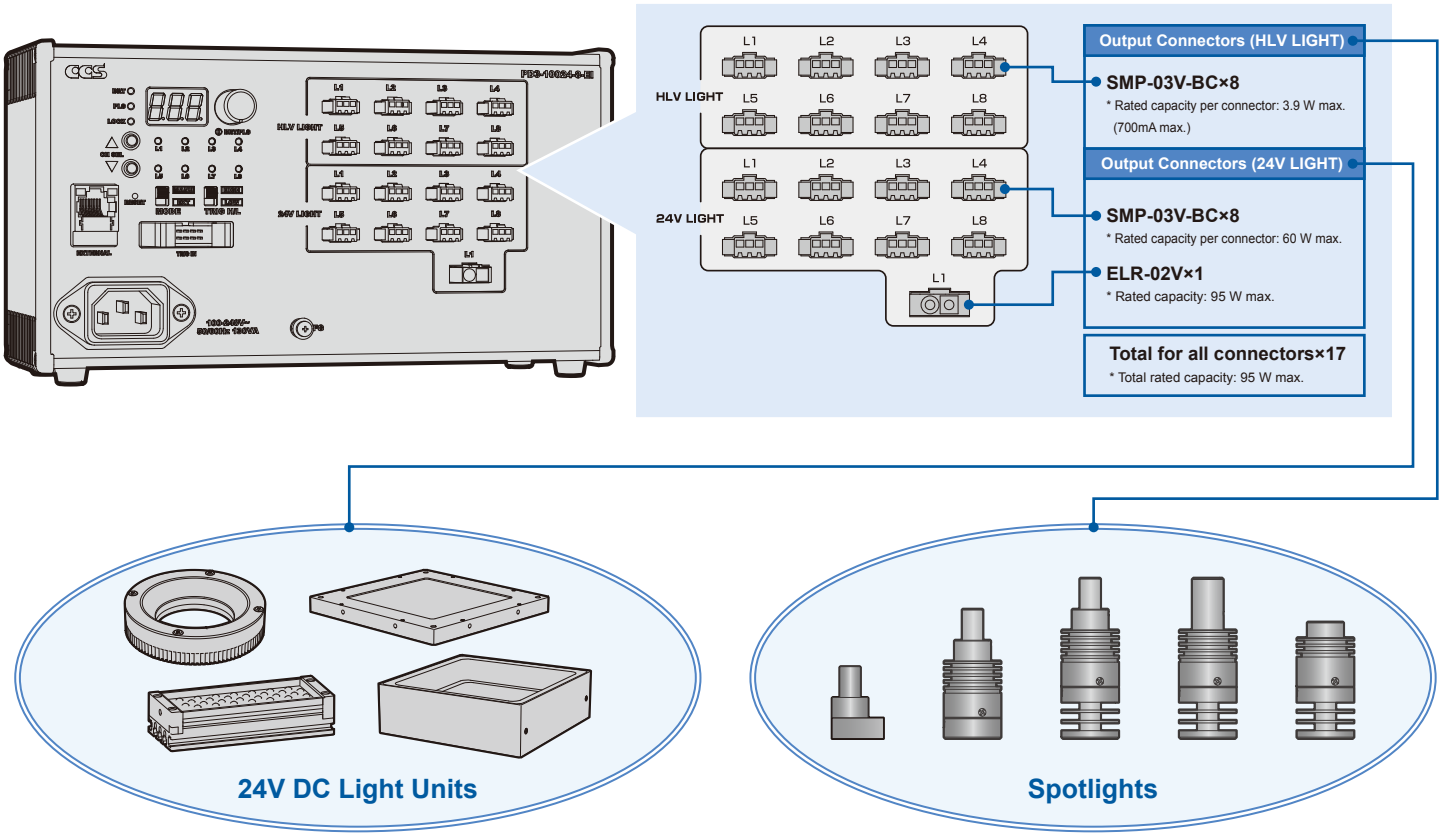
Select the control mode and lighting mode from the following Application Guide and proceed to the indicated reference items.

Application Guide

Lighting Mode \ Control Mode	MODE MANU EXT Front panel operation	MODE MANU EXT External control using a PLC or image process device
Continuous Mode	The Light Units are always ON. 	To use manual control in Continuous Mode, refer to items 1, 2, and 3 under **7 Manual Control**.	To use external control in Continuous Mode, refer to items 1, 2, and 3 under **8 Control with External Signals**. 
ON/OFF Mode	The Light Units are turned ON or OFF according to the external trigger signal input.  * It is possible to turn LED light unit ON and OFF by Ethernet communications, too.	To use manual control in ON/OFF Mode, refer to items 1, 2, and 3 under **7 Manual Control** and **9 Inputting the External Trigger**. 	External trigger signal or Ethernet communications setting can be selected when using ON/OFF mode by external control. To use external trigger signal in ON/OFF mode, refer to items 1, 2, and 3 under **8 Control with External Signals** and **9 Inputting the External Trigger**. To use Ethernet communications setting in ON/OFF mode, refer to items 1, 2, 3, and 5 under **8 Control with External Signals**. 
Strobe Mode	The Light Units are turned ON for a set time after the external trigger signal is input. 	To use manual control in Strobe Mode, refer to items 1, 2, 3 and 4 under **7 Manual Control** and **9 Inputting the External Trigger**. (Only 24V DC Light Units can be set.) 	To use external control in Strobe Mode, refer to items 1, 2, 3, and 4 under **8 Control with External Signals** and **9 Inputting the External Trigger**. (Only 24V DC Light Units can be set.) 

6 Light Unit Functions

This Control Unit can be connected to Light Units and Spotlights with 24V DC inputs. Connect 24V DC Light Units to the 24V LIGHT connectors and Spotlights to HLV LIGHT connectors. The functions vary with the Light Units that are connected. Check the following table before using the Light Units.



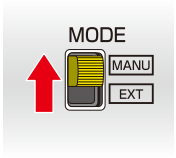
Item		24V LIGHT	HLV LIGHT	Reference page
Applicable illuminators		Light Units with 24V DC input voltage	Spotlights: HLV2 series, HLV series* *not including HLV-27 series/HLV-14-R/ HLV-14-GR/HLV-14-BL/HLV-14-SW	10
Lighting method		PWM control or lighting time control	Variable current control	10
Lighting mode	Continuous mode	○	○	5, 6
	ON/OFF mode	○	○	5, 6
	Strobe mode	○	---	5, 6
Control mode	Manual control	○	○	5
	External control	○	○	6, 7
Number of connectors	SMP-03V-BC	8	8	2
	ELR-02V	1	---	2
Rated capacity	SMP-03V-BC	Per connector: 60 W max.	Per connector: 3.9 W max. (700mA max.)	10
	ELR-02V	95W max.	---	10
Lighting delay time		Depends on the power consumption of the Light Units.	Depends on the light intensity of the Light Unit.	8
Channel selection		Depends on which Light Units are connected/disconnected.	Only connected Light Units can be selected.	5
Light Unit connection detection		Detected when connected for the first time.	Detected at any time.	5
Power startup time		0.5 s	3 s	10

- Make sure that the main power source is turned ON.
- Set items 1, 2, and 3 when using Continuous Mode or ON/OFF Mode.*
- Set items 1, 2, 3, and 4 when using Strobe Mode.

* If you have changed the lighting mode from the default value, set it to "Continuous Mode or ON/OFF Mode" in item 4.

1 Setting the Manual/External Mode Selector to Manual

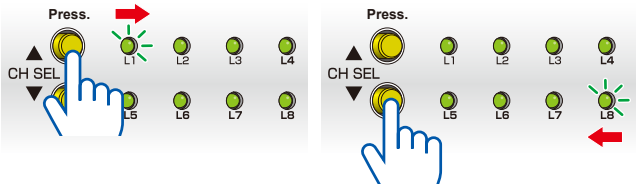
Set the Manual/External Mode Selector to MANU to set Manual Mode.



Check Make sure that the LOCK setting indicator is not light and that the trigger logic switch is set to HIGH. Otherwise you may not be able to perform the rest of this procedure.

2 Selecting the Channel

Press the channel selection switch to select the channel to set (L1 to L8).
Press the channel selection switch ▲, to select the Light Units in order from L1 to L8.
Press the channel selection switch ▼, to select the Light Units in order from L8 to L1.
Only channels with Light Units connected to them can be selected.
(If a new Light Unit is connected, the lowest channel is automatically selected.)



Eight channels from L1 to L8 are allocated to the 24V LIGHT and HLV LIGHT output connectors. (The ELR-02V connector for 24V LIGHT is L1.)
When L1 is selected, settings for the L1 Light Unit for the 24V LIGHT connectors (both SMP-03V-BC/ELR-02V) and the HLV LIGHT connectors can be changed.
The 8 channels can be controlled separately.

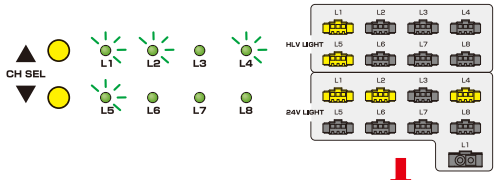
Precautions for Channel Selection

Only channels with Light Units connected to them can be selected. If a Light Unit is removed without turning OFF the power supply, the channel for the Light Unit that is no longer connected may be selected. This does not indicate a malfunction. Check the following table for details. There is risk of fire or electric shock. Make sure that the power supply is turned OFF when you connect Light Units or Spotlights.

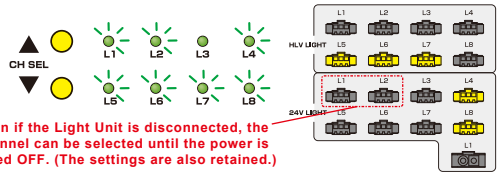
Operation	24V LIGHT	HLV LIGHT
The Light Unit was disconnected without turning OFF the power supply.	The channel with the disconnected Light Unit can be selected.	The channel with the disconnected Light Unit cannot be selected.
A Light Unit that is not supported was connected.	If there is an electrical connection, the Light Unit will light. (The channel can be selected.)	The Light Unit does not turn ON or an HLV ID error occurs. (The channel cannot be selected.)

Connector Connection Example

Channels L1, L2, L4 and L5 can be selected for the following connections:
HLV LIGHT : L1 and L5 and 24V LIGHT : L1, L2 and L4.



Channels L1, L2, L4, L5, L6, L7 and L8 can be selected if the connections are changed to the following:
HLV LIGHT : L5, L6, and L7 and 24V LIGHT : L4 and L8 without turning OFF the power supply.



Even if the Light Unit is disconnected, the channel can be selected until the power is turned OFF. (The settings are also retained.)

3 Setting the Light Intensity

Press the setting switch to light the BRT setting indicator.
Turn the setting switch to set a value between 0 and 255.
(Default setting: 000, Minimum: 000, Maximum: 255)



Digital display	Light Intensity (%)	
	24V LIGHT	HLV LIGHT
000	0.4 (Dimly lit)	0.0 (Not lit)
001	0.8	0.4
002	1.2	0.8
⋮	⋮	⋮
254	99.6	99.6
255	100.0	100.0

* The light intensities are theoretical values.
* When the small value larger than or equal to 001 is set to the Light Unit connected to the HLV LIGHT connector, depending on the model of the Light Unit and the cable length, the Light Unit may not turn ON.

4 Selecting the Lighting Mode

Press the setting switch to light the PLS setting indicator.
Turn the setting switch to select the lighting mode from Continuous Mode, ON/OFF Mode, or Strobe Mode. (Default value: F00)



4 Selecting the Lighting Mode (Continued)

Continuous Mode
Turn the setting switch and set F00 to turn ON the Light Units continuously.

ON/OFF Mode (If the external trigger is not used, the Light Units are ON continuously.)
Turn the setting switch and set F00 to turn the Light Units ON and OFF.
The Light Units are turned ON or OFF according to the external trigger signal input.

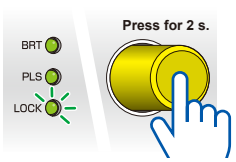
Strobe Mode (If an external trigger is not used, the Light Units are OFF.)
To flash the strobe, turn the setting switch and select a setting from F01 to F10 (strobe time of 40 µs to 40 ms). The Light Units are turned ON for the period of time set on the setting switch after the external trigger signal is input.
The Strobe Mode can be set for 24V DC Light Units only.

Digital display	Status	
F00	Continuous Mode or ON/OFF Mode	
F01	Strobe Mode (The Strobe Mode can be set for 24V DC Light Units only.)	40µs
F02		80µs
F03		120µs
F04		200µs
F05		600µs
F06		1ms
F07		4ms
F08		10ms
F09		20ms
F10		40ms

For details on the external trigger input, refer to 9. Inputting the External Trigger.

Locking Settings

When the setting switch is pressed for 2 seconds or longer, the lighting mode and light intensity settings are locked, and the LOCK setting indicator lights. (The set values can be viewed.) Pressing the switch again for 2 seconds or longer releases the lock.



8 Control with External Signals

An external device such as a PLC or image processing device transmits the send data to the Control Unit. The Control Unit processes the data and returns the results. The external device gets the receive data as the execution results.

Communications Specifications (*1)

TCP/IP protocol or UDP/IP protocol (Switching operation is not required.), and Ethernet (Baud rate: 10 Mbps or 100 Mbps, automatically detected; Transmission medium: 10BASE-T or 100BASE-TX)

Command Formats

Sample of Alphanumeric Characters: **ABCDEFGHIJKLMNOPQRSTUVWXYZ 0123456789**

Send Data (*2)

Function	Header	Channel specification	Sent command		Checksum	Delimiter	Default (*9)
			Instruction	Data (*3)			
Light Intensity Setting	@	00 to 07 (Refer to *4) FF: All channels (F instruction used for the batch setting for each channel and L instruction only)	F	000 to 255 (Refer to *5)	00 to FF (Refer to *8)	<CR><LF>	000
Lighting Mode Setting			S	00 to 10 (Refer to *6)			00
ON/OFF Setting			L	0: Not lit, 1: Lit (Refer to *7)			1
Setting Status Check			M	- - -			- - -
Error Status Check		00 (fixed)	C	- - -			- - -
All Channel Initialization			R	- - -			- - -
IP Address			E01	000.000.000.000 to 255.255.255.255 (Specify all digits, e.g., specify "192.168.000.005" instead of "192.168.0.5.")			192.168.000.002
Subnet Mask			E02				255.255.255.000
Default Gateway			E03				192.168.000.001
Reply IP Address			E05				192.168.000.016
Reception Port Setting			E04	00000 to 65535 (Specify all digits, e.g., specify "04561" instead of "4561.")			40001
Reply Port Setting			E06				30001

Receive Data (*10)

Function	Header	Channel specification	Received command				Checksum	Delimiter
			OK		NG			
Light Intensity Setting	@	00 to 07 (Refer to *4) FF: All channels (F instruction used for the batch setting for each channel and L instruction only)	O	- - -	N	01: Command error 02: Checksum error 03: Set value out of range error	00 to FF (Refer to *8)	<CR><LF>
Lighting Mode Setting								
ON/OFF Setting								
Setting Status Check				F999.S99.L9 (Refer to *11)				
Error Status Check		00 (fixed)		00: Normal, 11: Error				
All Channel Initialization				- - -				
IP Address								
Subnet Mask								
Default Gateway								
Reply IP Address								
Reception Port Setting								
Reply Port Setting								

<< Annotation>>

*1) The number of TCP connection (possible numbers to connect at same time) which PD3 correspond is "1"

*2) Send a data within 4 seconds from 'Header' to 'Delimiter', otherwise time-out error occurs and command data will be rejected.

*3) Specify all numbers in decimal format.

*4) Channel Specification

Eight channels from L1 to L8 are allocated to the 24V LIGHT and HLV LIGHT output connectors. (The ELR-02V connector for 24V LIGHT is L1.) When L1 is selected, settings for the L1 Light Unit for the 24V LIGHT connectors (both SMP-03V-BC/ELR-02V) and the HLV LIGHT connectors can be changed. The 8 channels can be controlled separately.

Channel	L1	L2	L3	L4	L5	L6	L7	L8
Set value	00	01	02	03	04	05	06	07

(Set values that are higher than 07 are not valid.)

*5) Light Intensity Settings

The light intensity is controlled to any of 256 levels. 000 to 255 (000: Minimum, 255: Maximum)

Light intensity (%)	24V LIGHT	0.4 (Dimly lit)	0.8	1.2	· · ·	99.6	100.0
	HLV LIGHT	0.0 (Not lit)	0.4	0.8	· · ·	99.6	100.0
Digital display		000	001	002	· · ·	254	255
Set value		000	001	002	· · ·	254	255

* The light intensities are theoretical values.
* When the small value larger than or equal to 001 is set to the Light Unit connected to the HLV LIGHT connector, depending on the model of the Light Unit and the cable length, the Light Unit may not turn ON.

*6) Lighting Mode Settings

Select the lighting mode form Continuous Mode, ON/OFF Mode, or Strobe Mode. The lighting time can be set in Strobe Mode. (The Strobe Mode can be set for 24V DC Light Units only.) For details on the external trigger input, refer to 9. Inputting the External Trigger.

Status	Continuous Mode or ON/OFF Mode	Strobe Mode									
		40µs	80µs	120µs	200µs	600µs	1ms	4ms	10ms	20ms	40ms
Digital display	F00	F01	F02	F03	F04	F05	F06	F07	F08	F09	F10
Set value	00	01	02	03	04	05	06	07	08	09	10

(Set values that are higher than 10 are not valid.)

Continuous Mode

Set the value to 00 to enter Normal Mode and keep the Light Unit lit continuously.

ON/OFF Mode

(If the external trigger is not used, the Light Units are ON continuously.)

Set the value to 00 to enter ON/OFF Mode and turn the Light Unit ON and OFF. The Light Units are turned ON or OFF according to the external trigger signal input.

Strobe Mode

(If an external trigger is not used, the Light Units are OFF.)

Select the lighting time from 01 to 10 (40 µs to 40 ms) to use a strobe light. The Light Units are turned ON for the set time after the external trigger signal is input. The Strobe Mode can be set for 24V DC Light Units only.

*7) ON/OFF setting from Ethernet communications without regards to trigger logic switch, turned OFF at '0' and ON at '1'.
When operating Ethernet communications and trigger signal input at same time in ON/OFF mode.
When Trigger logic switch is at HIGH: if either controls setting to OFF setting, Light unit will be turned OFF
When Trigger logic switch is at LOW: if either controls setting to ON setting, Light unit will be turned ON

*8) Checksum
The codes of the ASCII characters from the header to the send command are added, the lowest byte is converted to hexadecimal, and two characters are sent.
(* The lowest byte (two characters) of 17F is taken, so the checksum is 7F.)

Example: Setting the Light Intensity of Channel 2 to 125

	Header	Channel		Sent command				Total
	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	
Character	@	0	1	F	1	2	5	
ASCII (hexadecimal)	40 hex	30 hex	31 hex	46 hex	31 hex	32 hex	35 hex	17F hex

*9) Except for the ON/OFF setting, any changes to the default values will be held after the power supply is turned OFF. The ON/OFF setting will not be held.

*10) There is no received data when timeout error occurs.

*11) Received Command for Setting Status Check (F999.S99.L9): F999 = F instruction set value (F000 to F255), S99 = S instruction set value (S00 to S10), L9 = L instruction set value (L0: Not lit, L1: Lit) L1 must be returned immediately after the power supply is turned ON.

Batch Setting for Each Channel (Available Only When "FF" Is Set for the Channel Specification)
When you specify the data in the form "aaa/bbb/ccc/ddd/eee/fff/ggg/hhh" to the F instruction, you can set the light intensities for each channel with a single send data at a time. If you specify "FFF" for a data, the light intensity of the corresponding channel will not be changed.
Example) @FFF123/045/FFF/099/099/099/099/09986CRLF: Setting the light intensity to 123 for L1, 045 for L2. The light intensity for L3 will not be changed. Setting the light intensities to 099 for L4 to L8.
When you specify the data in the form "a/b/c/e/f/g/h" to the L instruction, you can set ON/OFF signals for each channel with a single send data at a time. If you specify "F" for a data, the ON/OFF status of the corresponding channel will not be changed.
Example) @FFL1/F/0/0/0/0/0/0F8CRLF: Setting the signal to ON for L1. The ON/OFF status for L2 will not be changed. Setting the signals to OFF for L3 to L8.

Setting Procedures

- Make sure that the main power source is turned ON.
- Set items 1, 2, and 3 when using Continuous Mode.*
- Set items 1, 2, 3, and 5 when using ON/OFF Mode.*
- Set items 1, 2, 3, and 4 when using Strobe Mode.

* If you have changed the lighting mode from the default value, set it to "Continuous Mode or ON/OFF Mode" in item 4.

1 Setting the Manual/External Mode Selector to External

Set the Manual/External Mode Selector to EXT to set External Mode. The value set with external control is displayed on the digital display.
* Although it is possible to set the external control setting even the manual/external mode selector is set to manual mode, the setting will not be activated until the manual/external mode selector is set to external mode.



2 Setting Up the Network (Only Initially and When Settings Are Changed)

Set the Unit's IP address and the reply address. To enable the settings that were changed, cycle the power supply.
* If the IP address changes, do not forget to change the send destination of commands.

Item	Setting example	Send data	Receive data when OK	Receive data when NG
IP Address	192.168.3.2	@00E01192.168.003.00230CRLF	@00OEFCRLF	@00N014FCRLF (when there is a command error) @00N0351CRLF (when there is a set value out of range error)
Subnet Mask	255.255.255.0	@00E02255.255.255.00035CRLF		
Default Gateway	192.168.3.1	@00E03192.168.003.00131CRLF		
Reception Port Setting	4561	@00E040456149CRLF		
Reply IP Address	192.168.3.10	@00E05192.168.003.01033CRLF		
Reply Port Setting	4562	@00E06045624CCRLF		

3 Setting the Light Intensity

Specify the channel and set the light intensity.

Setting example	Send data	Receive data when OK	Receive data when NG
Setting the L3 light intensity to 75	@02F07584CRLF	@02OF1CRLF	@02N0151CRLF (when there is a command error)

4 Setting the Lighting Mode

Specify the channel and set the lighting mode.

Setting example	Send data	Receive data when OK	Receive data when NG
Setting the L2 lighting mode to 200 µs in Strobe Mode	@01S0458CRLF	@01OF0CRLF	@01N0352CRLF (when there is a set value out of range error)

5 To Set ON/OFF Signal in ON/OFF Mode

Specify the channel and set ON/OFF signal.

Setting example	Send data	Receive data when OK	Receive data when NG
To turn all light units OFF	@FFL048CRLF	@FFO1BCRLF	@FFN027CCRLF (When there is checksum error)

▶ To Check the Setting Status

When checking the setting status, send the following command after specify the channel.

Setting example	Send data	Receive data when OK	Receive data when NG
To check the setting status of L1.	@00MEDCRLF	@00OF075.S04.L060CRLF	@00N0250CRLF

▶ Checking the Unit Status

Send the following command to check the Unit status.

Setting	Send data	Receive data when OK	Receive data when NG
Checking the Unit status	@00CE3CRLF	@00O004FCRLF (normal)	@00N0351CRLF (when there is a set value out of range error)

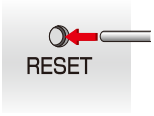
▶ Resetting the Light Intensity and Lighting Mode

To restore the external control setting to default value, send the following command.

Setting	Send data	Receive data when OK	Receive data when NG
All Channel Initialization	@00RF2CRLF	@00OEFCRLF	@00N014FCRLF (when there is a command error)

▶ Initialize the Network Setting

If the IP address setting and others is incorrect, you will not be able to connect to the Control Unit. You will also not be able to reset the IP address and others. If that occurs, use a pointed object to press the external control reset switch on the front panel to reset network settings to their default values. To enable the reset settings, cycle the power supply



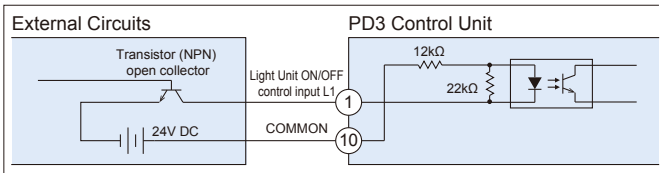
Input Signal and Photocoupler

The input signal from the external trigger input connector can be used to control the photocoupler inside the Unit to turn the LED Light Units ON and OFF or to control strobe timing. The operation depends on the setting of the trigger logic switch.

Trigger Logic Switch	Input signal	Photocoupler	ON/OFF Mode	Strobe Mode
HIGH	HIGH	OFF	Light Units ON	Light Units ON for the set time.
	LOW	ON	Light Units OFF	No change
LOW	HIGH	OFF	Light Units OFF	No change
	LOW	ON	Light Units ON	Light Units ON for the set time.

* When operating trigger signal input and Ethernet communications at same time in ON/OFF mode.
When Trigger logic switch is at HIGH: if either controls setting to OFF setting, Light unit will be turned OFF
When Trigger logic switch is at LOW: if either controls setting to ON setting, Light unit will be turned ON

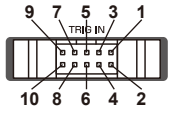
External Trigger Signal Connection Example



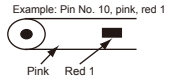
Signal Specifications					
Rated input voltage	Maximum input voltage	ON voltage/ ON current	OFF voltage/ OFF current	Lighting delay time	Input impedance
24V DC	26.4V DC	14.4V DC min./ 3 mA max.	5V DC max./ 1 mA max.	20us max. (24V LIGHT) 5ms max. (HLV LIGHT)	12 kΩ (per terminal)

Connector Layout

External Trigger Input Connector Plug



Enlarged Cable Diagram



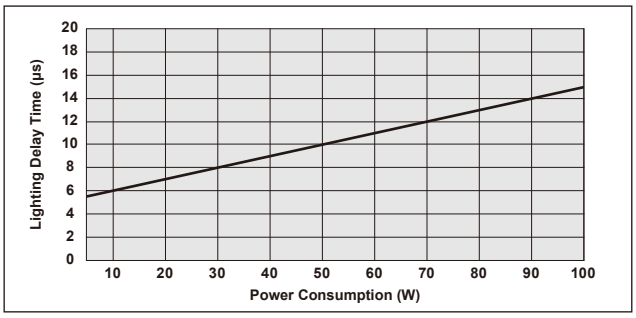
No.	Signal	Cable (EXCB2-M10-3)	
		Wire color	Marks
1	Light Unit ON/OFF control input (L1)	Orange	Black 1
2	Light Unit ON/OFF control input (L2)	Orange	Red 1
3	Light Unit ON/OFF control input (L3)	Gray	Black 1
4	Light Unit ON/OFF control input (L4)	Gray	Red 1
5	Light Unit ON/OFF control input (L5)	White	Black 1
6	Light Unit ON/OFF control input (L6)	White	Red 1
7	Light Unit ON/OFF control input (L7)	Yellow	Black 1
8	Light Unit ON/OFF control input (L8)	Yellow	Red 1
9	Not used.	Pink	Black 1
10	COMMON+	Pink	Red 1

Lighting Delay Time

The lighting delay time for lights connected to 24V LIGHT connectors depends on the power consumption of the Light Unit. The lighting delay time for Spotlights connected to HLV LIGHT depends on the intensity of the Spotlights. Refer to the following graphs and tables.

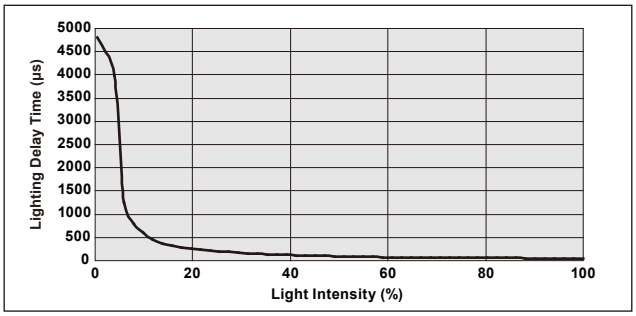
* The data in the graphs show reference values when a Light Unit or Spotlight with a 5-m cable is used. (The values are for reference only.)

Power Consumption vs. Lighting Delay Time Characteristic (24V LIGHT)



Power Consumption	10W	20W	30W	40W	50W	60W	70W	80W	90W	95W
Lighting Delay Time (typ.)	6µs	7µs	8µs	9µs	10µs	11µs	12µs	13µs	14µs	14.5µs

Light Intensity vs. Lighting Delay Time Characteristic (HLV LIGHT)



Light Intensity	10%	20%	30%	40%	50%	60%	70%	80%	90%	100%
Lighting Delay Time (typ.)	610µs	260µs	180µs	130µs	110µs	82µs	73µs	65µs	60µs	55µs

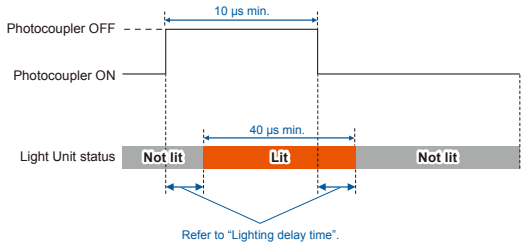
Trigger Input Sequence Diagram

- A pulse width of ON signal shall be 10µs or more. The Light Units will be truned on for at least 40µs, even when the input ON signal is less than 40µs.
- If another trigger is input before the Light Unit turns OFF in Strobe Mode, the starting point of the reentered trigger is taken as the start time and the strobe light continues for the set time from that point.

ON/OFF Mode

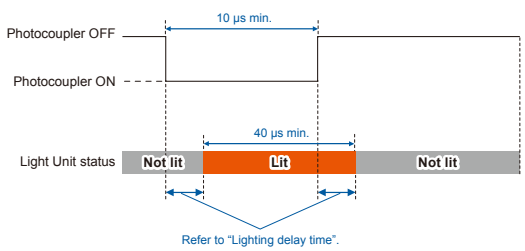
● Trigger Logic Switch Set to HIGH
The Light Units turn ON when the photocoupler is OFF, and OFF when the photocoupler is ON.





● Trigger Logic Switch Set to LOW
The Light Units turn ON when the photocoupler is ON, and OFF when the photocoupler is OFF.

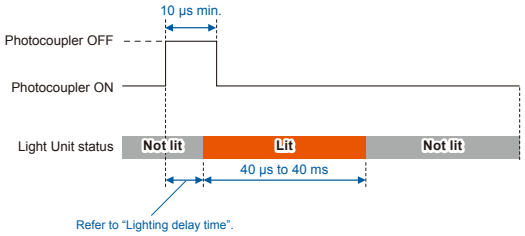




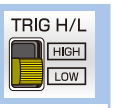
Strobe Mode (Only 24V DC Light Units can be set.)

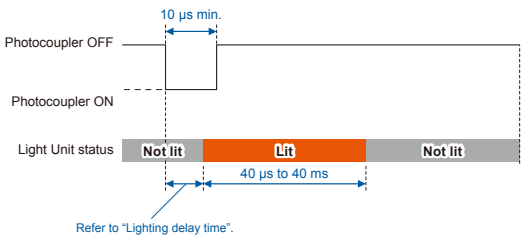
● Trigger Logic Switch Set to HIGH
From the point where the photocoupler goes OFF, the Light Units are turned ON for the set time (40 µs to 40 ms).





● Trigger Logic Switch Set to LOW
From the point where the photocoupler goes ON, the Light Units are turned ON for the set time (40 µs to 40 ms).





10 Errors

If an internal error occurs in the Control Unit, it is displayed on the digital display. Refer to the following table and clear the error before using the Control Unit again.

Digital display	Error	Status	Clearing the Error	Recovery Method
oCP	Overcurrent Error The current consumption of the Light Unit exceeded 107% of the rating.	Output is stopped.	Check the rating of the LED Light Unit. Connect an LED Light Unit that is within the rating.	Press and hold the setting switch or cycle the power.
EFn	Fan Stop Error The fan has stopped.	Output is stopped.	Doing so may cause product failure. Consult a CCS representative.	Press and hold the setting switch or cycle the power.
Eld OR 125 (Example display) Period	HLV ID Error A Spotlight outside the ID range was connected to an HLV LIGHT connector. If the channel that is connected to a Spotlight that is outside the ID range is selected, an error will be displayed. Eld If no Light Unit has ever been connected to the same channel in the 24V LIGHT connectors, Eld will be displayed. 125 If a Light Unit is connected to the same channel in the 24V LIGHT connectors, "□□□" is displayed. ("□□□" is the set value.) Light Units that are connected to 24V LIGHT connectors can be set normally.	Normal	Check the Spotlight. Connect a Spotlight that is supported by the Control Unit.	Operation recovers automatically.

Error Output

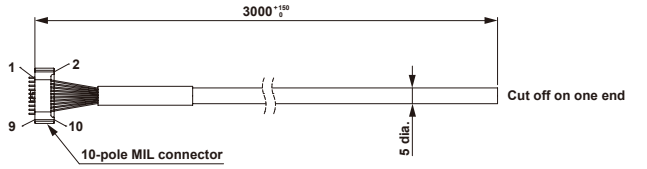
When detecting the error during external control, command will be received as acknowledgement for checking status (over current confirmation) command "C". (This applies only to overcurrent errors and fan stop errors.)

* Immediately after the error, occurrence of an error will be noticed only one time by using UDP protocol. Notify data is the same as checking status (over current confirmation) command "C".

For details, refer to 8. Control with External Signals.

11 Optional Accessories (Sold Separately)

External Trigger Input Cable — Model: EXCB2-M10-3



Connector: XG4M-1030-T (manufactured by OMRON)

12 Troubleshooting

If you have any problems during product usage, please look up the cause in this chart. If the situation does not improve, or an unexpected situation occurs, please contact CCS Inc.

Symptom	Items to check for fixing the problem	Reference page
Lights not illuminating.	Are all power sources turned ON?	3
	Is the LED light cable inserted firmly into the output connector?	3
	Is the AC cord inserted firmly into the AC inlet and a wall socket?	3
	Does the digital display show an error? Check the error display.	9
	Is an appropriate Light Unit connected to the output connector? Check the Light Unit.	4
	Is the output connector the correct one for the connected Light Unit? Connect 24V DC Light Units to the 24V LIGHT connectors and Spotlights to HLV LIGHT connectors.	4
	Is the lighting mode set correctly? Set the lighting mode to F00 for Continuous Mode or ON/OFF Mode.	5, 6
	Does the logic of the trigger signal match the trigger signal setting switch?	8
	Is the external trigger input OFF (in ON/OFF Mode or Strobe Mode)? Check the setting method and sequence diagram.	8
Light intensity control is not possible.	Is the manual/external switch set correctly? Set it to MANU to operate manually and to EXT for external control.	5, 6
	Is an appropriate Light Unit connected to the output connector? Check the Light Unit.	4
	Is the output connector the correct one for the connected Light Unit? Connect 24V DC Light Units to the 24V LIGHT connectors and Spotlights to HLV LIGHT connectors.	4
	Have the wrong channels been selected? Check the channels whose lights are to be controlled.	5, 6
	Is the setting switch locked? Press and hold the setting switch for more than two seconds to release the lock.	5

Symptom	Items to check for fixing the problem	Reference page
External control is not possible.	Is the external control cable (LAN cable) inserted firmly into the external control connector and an external device?	3
	Was the command sent correctly? Set the network settings and data correctly.	6, 7
Unable to turn the Light Unit ON and OFF or use strobe control.	Is the external trigger input cable inserted firmly into the external trigger input connector and an external device?	3
	Are the external trigger input cable and external devices wired correctly? Check the connector configuration and correct the wiring.	8
	Is the lighting mode set correctly? Set the value to F00 for ON/OFF Mode. Set the value to F01 to F10 for Strobe Mode.	5, 6
	Is the signal setting correct? Check the setting method and sequence diagram.	8
Malfunctioning.	Please use designated power sources with stable voltage. Sharing power sources with inverters, motors, etc., may cause malfunction.	–
	Do not bundle product cables with high-voltage lines or power lines. Doing so may cause the product to malfunction. Keep the product cables as far away from such lines as possible.	–
Fuming, extreme temperature, smell, noise, or other abnormality.	There is a possibility of product failure. Please stop usage immediately and turn OFF the power switch. Please do not attempt to use or repair the product, since it is dangerous, but contact CCS Inc.	–

13 Main Specifications

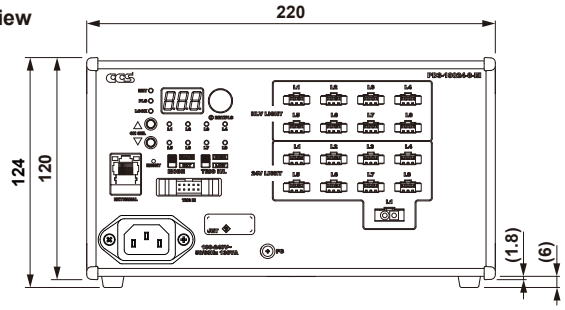
Product name	Digital Control Unit for LED Light Units (with Ethernet communications)
Model	PD3-10024-8-EI(A)
Rated capacity	95 W max. for 17 connectors total
Input power supply	100 to 240V AC (+10%, −15%), 130 VA, 50/60 Hz
Inrush current (typ.)	15 A (at 100V AC), 30 A (at 200V AC) from a cold start
Ground leakage current	3.5 mA max. (264V AC, 60 Hz, with no load)
Rated output voltage	24V DC
Insulation withstand voltage (input-output, input-FG)	1,500V AC for one minute, Cutoff current: 10 mA, 500V DC, 20 MΩ min.
Operating temperature and humidity	Temperature: 0 to 40°C, Humidity: 20% to 85% (with no condensation)
Storage temperature and humidity	Temperature: −20 to 60°C, Humidity: 20% to 85% (with no condensation)
Vibration resistance	Acceleration: 19.6 m/s², Frequency: 10 to 55 Hz, Cycles: 3 minutes, Sweep cycle: For 1 hour each in X, Y, and Z directions
Cooling method	Forced air cooling
CE Marking	Safety standard: Conforms to EN 61010-1, EMC standard: Conforms to EN 61326, Class A.
PSE	Specified Electrical Appliance and Material (DC power supply units), Conforms to Technical Standards
Environmental regulations	RoHS compliant
Input connector	AC input: 3-pin inlet EN 60320-1 certified C14 type × 1
External control connector	Trigger input: MIL connector (MIL-C-83503 compliant), 10-pole For setting the light intensity and lighting mode: RJ-45
Material and surface processing	Material: Aluminum and resin, Surface processing: Blue alumite
Weight	1500 g max.
Accessories	2-m long 3-pin power cord with ground terminal x1, Base Brackets x1 set, Instruction Guide x1

Specifications for Different Output Connectors

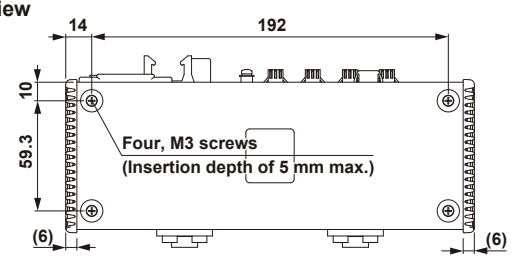
Input type	24V LIGHT	HLV LIGHT
Applicable illuminators	24V DC light units	Spotlights: HLV2 series, HLV series* *not including HLV-27 series/HLV-14-R/ HLV-14-GR/HLV-14-BL/HLV-14-SW
Rated capacity	60W max. (SMP-03V-BC/1 connector) 95W max. (ELR-02V)	3.9W (700mA) max. (SMP-03V-BC/1 connector)
Lighting method	PWM control (125kHz) or lighting time control	Variable current control
Light Unit connection detection	Detected when connected for the first time.	Detected at any time.
Power startup time	0.5 s	3 s
Output connectors	SMP-03V-BC (J.S.T. Mfg. Co., Ltd.) x 8 ELR-02V (J.S.T. Mfg. Co., Ltd.) x 1	SMP-03V-BC (J.S.T. Mfg. Co., Ltd.) x 8

14 Dimensions

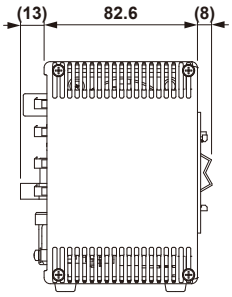
Front View



Bottom View



Side View



Environmental Regulation

EU RoHS Directive

The RoHS Directive is short for the "restriction of use of certain hazardous substances in electrical and electronic equipment." As a directive, it restricts the use of specific hazardous substances for new electrical and electronic equipment marketed in the EU on or after July 1, 2006, and restricts the use of six substances, which are (1) lead, (2) mercury, (3) cadmium, (4) hexavalent chromium, (5) polybrominated biphenyl (PBB), and (6) polybrominated diphenyl ether (PBDE).

*Standards for "RoHS Directive-Compliant Products"

Lead	Mercury	Cadmium	Hexavalent chromium	PBB	PBDE
1000 ppm max.	1000 ppm max.	100 ppm max.	1000 ppm max.	1000 ppm max.	1000 ppm max.

(Items that are exempted in the RoHS Directive are excluded from these standards.)

China RoHS Directive

China RoHS Directive is formally known as "Management Methods for Controlling Pollution by Electronic Information Products", which was implemented on March 1, 2007 in China. Same as EU RoHS Directive, this regulation restricts the usage of six substances such as lead, mercury, cadmium, hexavalent chromium, polybrominated biphenyl (PBB), and polybrominated diphenyl ether (PBDE). This regulation requires electronic information products which are manufactured or imported, and sold in China, to clearly disclose contents of the 6 restricted substances listed below.

Name and Amount of Toxic and Hazardous Substances or Elements

Usage deadline for environmental protection	Product name	Toxic or hazardous substances and elements					
		Lead (Pb)	Mercury (Hg)	Cadmium (Cd)	Hexavalent chromium (Cr (VI))	PBB	PBDE
10	Control Unit for LED Lights	×	○	×	○	○	○

○ : Indicates that this toxic or hazardous substances contained in all the homogeneous materials for this part, according to SJ/T11363-2006 is within the limit requirement.

× : Indicates that this toxic or hazardous substance contained in all the homogeneous materials for this part, according to SJ/T11363-2006, is over the limit requirement.

*Lead and cadmium are excluded in EU RoHS.

Usage Deadline for Environmental Protection

The number used in this logo is based on "Management Methods for Controlling Pollution by Electronic Information Products" and related regulations from People's Republic of China. It shows the product usage duration in years for environmental protection. After finishing a product usage, the product needs to be re-used or discarded appropriately following local law and regulations, complying with safety and usage caution.

产品中有毒有害物质或元素的名称及含量

环保使用期限	产品	有毒有害物质或元素					
		铅 (Pb)	汞 (Hg)	镉 (Cd)	六价铬 (Cr (VI))	多溴联苯 (PBB)	多溴二苯醚 (PBDE)
10	LED 照明专用电源	×	○	×	○	○	○

○ : 表示该有毒有害物质在该部件所有均质材料中的含量均在 SJ/T11363-2006 标准规定的限量要求以下。

× : 表示该有毒有害物质至少在该部件的某一均质材料中的含量超出 SJ/T11363-2006 标准规定的限量要求。

(注) 铅和镉中的“×”，因欧洲 RoHS 没限定，故用“○”表示。

环保使用期限

此标志的数字是根据中华人民共和国电子信息产品污染控制管理办法以及有关标准等，表示该产品的环保使用期限的年数。
遵守产品的安全和使用上的注意，在产品使用后采取适当的方法根据各地法律，规定，回收再利用或进行废弃处理。

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WARRANTY PERIOD: TWO YEARS, STARTING FROM CCS Inc. SHIPPING DATE.

CCS Inc. WILL REPAIR OR REPLACE THE PRODUCT FREE OF CHARGE IF IT SHOULD FAIL TO FUNCTION WITHIN THE SPECIFIED WARRANTY PERIOD. IF EITHER OF THESE CONDITIONS OCCURS, PLEASE TAKE THE PRODUCT TO YOUR CCS SALES REPRESENTATIVE.

WARRANTY TERMS

1 CCS Inc. WILL REPAIR OR REPLACE THE PRODUCT FREE OF CHARGE IF IT SHOULD FAIL TO FUNCTION UNDER USE ON OUR SPECIFIED CONDITION IN ACCORDANCE WITH THE INSTRUCTION GUIDE AND OTHER WRITTEN CAUTIONS DURING THE INDICATED WARRANTY PERIOD OF TWO YEARS.

2 CCS Inc. WILL CHARGE A REPAIR FEE UNDER THE FOLLOWING CONDITIONS:
1) IF THE PRODUCT HAS BEEN SUBJECTED TO MISUSE, UNAUTHORIZED REPAIRS, OR MODIFICATION FROM ITS ORIGINAL DESIGN.
2) IF THE PRODUCT HAS BEEN DAMAGED FROM IMPACTS DUE TO INAPPROPRIATE HANDLING.
3) IF DAMAGE TO THE PRODUCT RESULTS FROM EXTERNAL CAUSES INCLUDING ACCIDENTS, FIRE, POLLUTION, RIOTS, COMMUNICATION FAILURES, EARTHQUAKES, THUNDERSTORMS, WIND AND FLOOD DAMAGE, OR ANY OTHER ACT OF PROVIDENCE, OR FROM ANY EXTRAORDINARY CONDITIONS SUCH AS ELECTRICAL SURGES, WATER LEAKAGE, CONDENSATION, OR THE USE OF CHEMICALS.
4) IF THE DAMAGE RESULTS FROM CONNECTION TO ANY LED LIGHT UNIT OR TO ANY EQUIPMENT WHICH CCS Inc. DOES NOT MANUFACTURE OR DOES NOT SPECIFY FOR USE.

3 CCS ASSUMES NO LIABILITY FOR ANY PURCHASER'S SECONDARY DAMAGE (DAMAGE OF EQUIPMENT, LOSS OF OPPORTUNITIES, LOSS OF PROFITS, ETC.) OR ANY OTHER DAMAGE RESULTING FROM A FAILURE OF OUR PRODUCT.

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Do not use the product in the following situations.

- Under conditions or in an environment not described in this instruction guide.
- In nuclear energy control systems, railroad systems, aviation systems, vehicles, combustion equipment, medical equipment, amusement machines, or safety equipment.
- In applications involving serious risk to life or property, particularly applications demanding a high level of safety.

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Ask any product queries to the following address or to your nearest CCS representative.

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